

Pair Pre-Screening: Analysis & Proposal

Prepared for: Stakeholder discussion **Author:** Lesandro (Lead) · 2026-06-19 **Status:** For discussion.

Recommendations are actionable; the numeric thresholds are deliberately left for stakeholder calibration.

Companion (technical): [docs/pair-prescreen-proposal.md](#) (design detail) · [scripts/pair_prescreen.py](#) (working POC)

Executive summary

A stakeholder proposed inserting a **lightweight Pair Screening step before full dashboard development**, so that pairs with limited practical value or poor operational feasibility do not consume the full dashboard-build + deep-QC cycle. This paper tests that idea against our existing portfolio of 13 live pairs.

Finding: the proposal is sound and **immediately feasible at near-zero added cost**. All four screening dimensions the stakeholder named map directly onto data we *already produce* in the Data and Econometrics stages — i.e. **before** the expensive charting, narrative, portal-assembly and deep-QC stages. A screen placed there can divert weak pairs early and roughly **halve the build cost of every pair that gets rejected**.

Recommendation: adopt the screen as a gate after the Econometrics stage, but calibrate its thresholds collaboratively — and, more importantly, act on two structural findings the screening exercise surfaced (below), which matter more than any individual pair verdict.

1. Background

Our original operating thesis was to build one high-quality dashboard (HY-IG) and replicate it cheaply across pairs. In practice, each pair has its own story, evidence and strategy logic, so each requires its own deep quality review — checking that Story, Evidence, Strategy and Methodology are mutually consistent, that the rank-1 strategy connects logically to the rest, that the analysis is persuasive, and that the economic logic holds. **That review is expensive, and today it happens only after a pair is fully built.**

The stakeholder's proposal: assess pairs at a higher level *first*, on four dimensions —

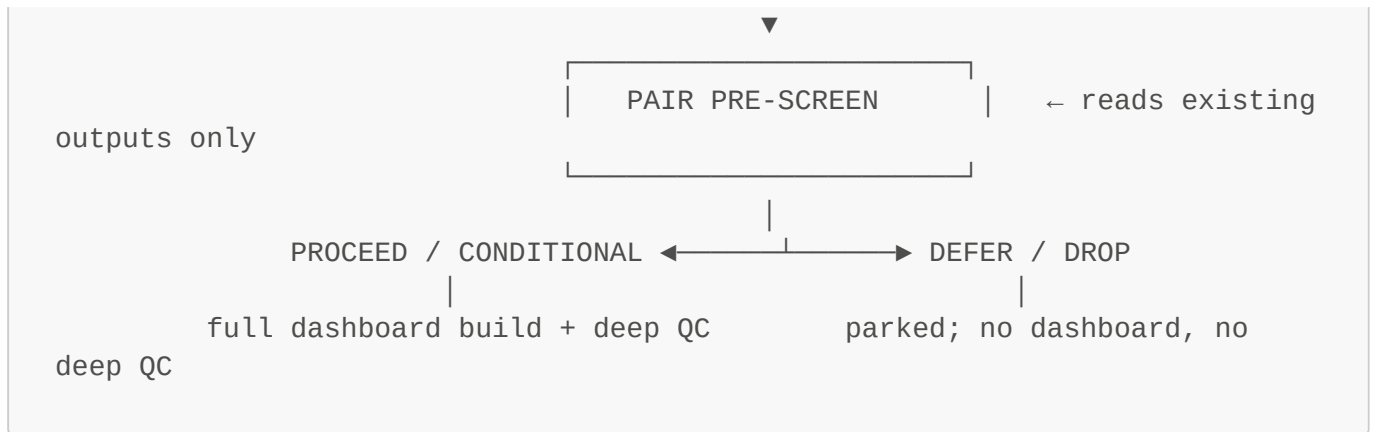
1. **Strategy Performance** — trade count, Sharpe, drawdown vs. benchmark;
2. **Operational Practicality** — data release delay, data availability, execution feasibility;
3. **Crisis Validation** — did the signal work across different market crises;
4. **Durability** — effective only in one period, or persistent across regimes.

— and only invest the full build + QC cycle in pairs that clear the bar.

2. The proposal in one picture

The key realisation is that the four dimensions are **already measured** by the time the Econometrics stage finishes. The screen therefore needs no new analysis — it reads existing outputs and renders a verdict:

Data sourcing → Data build → Econometrics + strategy tournament



Dimension	What we already produce that measures it
Strategy Performance	Out-of-sample Sharpe, uplift vs. buy-and-hold, drawdown reduction, trade count, turnover
Operational Practicality	Data source type & freshness, signal release lag, real-time tradability, turnover
Crisis Validation	Per-crisis strategy performance (Dot-Com, GFC, COVID, 2022)
Durability	In-sample vs. out-of-sample gap, statistical significance, winner-vs-field comparison, validation status

Verdict model: each dimension is scored Green / Amber / Red; the pair is then **PROCEED** (build), **CONDITIONAL** (build, but commit upfront to foregrounding the weak dimension), or **DEFER/DROP** (park, do not build). The screen does **not** replace deep QC — it decides *which pairs earn it*.

3. Analysis: what the portfolio tells us

We built a working scorecard and ran it across all 13 live pairs. With a first, deliberately rough threshold set, the screen produced a **credible separation that matched independent expert judgement**:

Verdict	Pairs	Why
DEFER/DROP	ism_services, phlxsox, gold_copper, vix_vix3m	ism_services: in-sample Sharpe -0.11 vs. out-of-sample 1.54 — a regime-lucky window, not a stable edge. The others: operationally heavy turnover (16–23×/year) that erodes practicality.
CONDITIONAL	busloans, hy_ig (×2), indpro (×2), m2sl, permit, petrol, umcsent	Real, useful edges with caveats. busloans is strongest: Sharpe 1.50, +0.61 over buy-and-hold, -1.0% drawdown vs. -23.9%, COVID-period Sharpe 2.75, low turnover.
PROCEED	(none)	See finding (a) below.

That the automated screen independently flagged the same pairs an analyst would is the central feasibility result: **the signal needed to triage pairs is already in our data**.

Two structural findings — more important than the verdicts

(a) Not one pair qualifies as fully "PROCEED" — because none has passed a true final exam. Every recent pair is labelled *found-in-search*: its headline performance comes from the same data used to *select* the strategy, with no untouched holdout test. The screen makes this portfolio-wide reality impossible to ignore. A genuine "PROCEED" verdict requires a **final-exam (holdout) step that does not yet exist in our pipeline.** This is the single most consequential gap the exercise revealed.

(b) Every pair shows a large jump from in-sample to out-of-sample Sharpe. This is structural: because the strategy tournament *selects* on out-of-sample performance, the winner's out-of-sample number is upward-biased. The practical implication for screening: **durability should be judged on the in-sample-to-out-of-sample gap, not the out-of-sample level alone.** A strong out-of-sample Sharpe sitting on a negative in-sample Sharpe (as in `ism_services`) is a red flag, not a green light.

A calibration caution

Our first threshold set was **too strict** — it would have dropped **bus loans**, one of our best pairs. The fix was straightforward (two thresholds were economically wrong), but the episode confirms the stakeholder's own caveat: **the framework is robust; the thresholds are where judgement and risk-appetite live, and they must be set with stakeholders, not by the tool.**

4. Recommendations

1. **Adopt the pre-screen as a gate after the Econometrics stage.** It is cheap (reads existing outputs), and it protects the most expensive part of the process (full build + deep QC).
2. **Calibrate thresholds collaboratively.** Use the working scorecard as a live instrument: run it on the existing 13 pairs, agree where the Green/Amber/Red lines sit, and encode the stakeholders' risk appetite rather than a developer's guess.
3. **Build the missing final-exam (holdout) step.** This is the highest-value structural improvement the exercise surfaced — without it, no pair can be more than provisionally trusted, and "Durability" can never be fully green. It also directly answers the stakeholder's "is it persistent across regimes?" question.
4. **Judge durability on the in-sample-to-out-of-sample gap,** not the out-of-sample headline. Make this an explicit screen rule.
5. **Generalise the operational and benchmark checks.** Two cheap upgrades materially strengthen the screen: (i) a uniform "does it beat a naive own-momentum benchmark?" test (already done for one pair) to catch leveraged-beta masquerading as alpha; (ii) capturing data-source freshness and release lag as structured fields so Operational Practicality scores without manual input.

5. Decision points for stakeholders

1. **Risk appetite / thresholds** — where should the Green/Amber/Red lines fall for Sharpe, uplift, turnover, crisis behaviour, and the in-sample/out-of-sample gap?
2. **Final-exam step** — do we commit to building the holdout test? (Recommended; it is the gating dependency for any real "PROCEED".)
3. **Treatment of already-built marginal pairs** — several live pairs would screen as DEFER/CONDITIONAL today. Do we retro-apply the screen (relabel/retire), or apply it only going forward?
4. **Granularity (Option D)** — the stakeholder leans Weekly + Monthly. The screen is granularity-agnostic, but Operational Practicality (release lag, tradability) interacts with frequency; we recommend aligning

the screen once Option D is settled.

5. **Gate strictness** — should DEFER be a hard stop, or a "review before proceeding" soft gate that a human can override?

6. Bottom line

The stakeholder's instinct is correct and the cost of acting on it is low. Most of the screening machinery already exists; what is missing is (1) agreed thresholds and (2) a true holdout final exam. The screening exercise has, as a by-product, told us something uncomfortable but valuable: **our current portfolio is entirely search-phase, and we have been judging strategies on numbers that are upward-biased by construction.** Pre-screening is worth adopting — but its greatest immediate value is having made that gap visible.

Appendix — POC scorecard (13 live pairs, illustrative thresholds)

Dimensions: **P** Strategy Performance · **O** Operational Practicality · **C** Crisis Validation · **D** Durability. (● strong · ● caveated · ● fails · ● insufficient data)

Pair	P	O	C	D	Verdict
busloans_spy	●	●	●	●	CONDITIONAL
gold_copper_xli	●	●	●	●	DEFER/DROP
hy_ig_spy	●	●	●	●	CONDITIONAL
hy_ig_v2_spy	●	●	●	●	CONDITIONAL
indpro_spy	●	●	●	●	CONDITIONAL
indpro_xlp	●	●	●	●	CONDITIONAL
ism_services_spy	●	●	●	●	DEFER/DROP
m2sl_yoy_spy	●	●	●	●	CONDITIONAL
permit_spy	●	●	●	●	CONDITIONAL
petrol_inv_spy	●	●	●	●	CONDITIONAL
phlxsox_spy	●	●	●	●	DEFER/DROP
umcsent_xlv	●	●	●	●	CONDITIONAL
vix_vix3m_spy	●	●	●	●	DEFER/DROP

Reproduce: [python scripts/pair_prescreen.py](#). Thresholds are illustrative and uncalibrated; the table demonstrates feasibility and separation, not final verdicts.